

B. Tech Degree V Semester Examination November 2010**ME 504 THERMAL ENGINEERING**

(2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART – A(Answer ALL questions)

(8 x 5 = 40)

- I. (a) Explain the reason of Poor part load thermal efficiency of the SI engine and CI engine.
- (b) Explain by means of suitable graphs the effect of dissociation on maximum temperature and brake power. How does the presence of Co effect dissociation?
- (c) With a neat sketch explain gear driven and exhaust driven supercharging methods.
- (d) Explain the various factors that influence the flame speed.
- (e) Define critical pressure ratio for the nozzle of the steam turbine. Obtain analytically its value in terms of the index of expansion.
- (f) Explain 'reheat factor'. Why is it's magnitude always greater than unity?
- (g) What is the effect of pressure ratio on the efficiency of a regenerative gas turbine cycle? At what pressure ratio the regenerator becomes superfluous?
- (h) What is the object of blade cooling in gas turbine and how it is done?

PART – B

(4 x 15 = 60)

- II. (a) The full load test on a two stroke engine yielded the following results : speed 440 rpm; brake load 50 Kg; imep 3 bar; fuel consumption 5.4 Kg/h; rise in jacket water temperature 36°C ; jacket water flow 440 Kg/h; air-fuel ratio by mass 30; temperature of exhaust gas 350°C ; temperature of the test room 17°C ; barometric pressure 76 cm of Hg; cylinder diameter 22 cm; stroke 25 cm; brake diameter 1.2 m; calorific value of fuel is 43 MJ/Kg; proportion of hydrogen by mass in the fuel 15%; $R = 0.287 \text{ kJ/Kg.K}$; specific heat of dry steam 2 kJ/Kg K. Assume enthalpy of superheated steam to be 3180 kJ/Kg. Determine,
- the indicated thermal efficiency
 - the specific fuel consumption in g/KWh
 - Volumetric efficiency based on atmospheric conditions.

Draw up a heat balance for the test on the percentage basis indicating the content of each item in the balance. (10)

- (b) How do the specific heats vary with temperature? Explain with the help of a PV diagram the loss due to variation of specific heats in an Otto cycle. (5)

OR

- III. (a) The compression ratio of an engine is 10 and the temperature and pressure at the start of compression is 37°C and 1 bar. The compression and expansion processes are both isentropic and the heat is rejected at exhaust at constant volume. The amount of heat added during the cycle is 2730 kJ/Kg. Determine the mean effective pressure and thermal efficiency of the cycle if
- the maximum pressure is limited to 70 bar and heat is added at both constant volume and constant pressure
 - all the heat is added at constant volume.

In this case how much additional work per Kg of charge would be obtained if it were possible to expand isentropically the exhaust gases to their original pressure of 1 bar. Assume that the charge has the same physical properties as that of air. (10)

- (b) By means of a Sankey diagram explain the energy flow through an engine. (5)

(P.T.O)

- IV. (a) Describe with a sketch the working of a free piston engine. (9)
 (b) Write short notes on –
 (i) After burning
 (ii) HUCR
 (iii) Fuel additives (3)
 (c) What is the function of gear box in automobile transmission system? (3)

OR

- V. (a) Discuss in detail about the governing of IC engines. (5)
 (b) Explain the function of fuel injector? What are the types of fuel injectors. (5)
 (c) Explain with figures various types of combustion chambers used in CI engines. (5)

- VI. The following data relate to a stage of an impulse reaction turbine.

Steam velocity coming out of nozzle	= 245 m/s
nozzle angle	= 20°
blade mean speed	= 145 m/s
speed of the rotor	= 300 rpm
blade height	= 10cm
specific volume of steam at nozzle outlet } and blade out let respectively	= 3.45 m³/Kg and 3.95 m³/Kg
power developed by the turbine	= 287 KW
efficiency of nozzle and blades combinedly	= 90%
carry over co-efficient	= 0.82

- Find (i) The heat drop in each stage.
 (ii) Degree of reaction
 (iii) Stage efficiency (15)

OR

- VII. (a) Justify the statement “Nozzles are more efficient than diffusers”. (3)

- (b) Derive the following expression for nozzle flow: $\frac{dA}{A} = \frac{1}{r} \frac{dp}{p} \left(\frac{1-M^2}{M^2} \right)$, the symbols having usual meanings. (6)

- (c) It is proposed to design steam nozzles for the following data:

Initial pressure	= 30 bar
Initial temperature	= 450°C
Back pressure	= 6 bar
Nozzle efficiency	= 90%
Initial steam velocity	= 60 m/s
Mass flow rate	= 2 Kg/s.

Assuming circular cross section, calculate the inlet, throat and exit diameter of nozzle. (6)

- VIII. In a closed cycle gas turbine plant the working fluid at 38°C is compressed with an adiabatic efficiency of 82%. It is then heated at constant pressure to 650°C. The fluid then expands down to initial pressure in a turbine with an adiabatic efficiency of 80%. The fluid after expansion is cooled to 38°C. The pressure ratio is such that work done per Kg is maximum. The working fluid is air which is assumed to be a perfect gas having $C_p = 0.25$ and $R = 29.2$ Kg.m/Kg.K. Calculate the pressure ratio and cycle efficiency. (15)

OR

- IX. (a) Explain surging and stalling in case of axial flow compressors. (5)
 (b) What are the limiting factors in turbine design? (5)
 (c) Write notes on –
 (i) Pressure loss combustion process and stability loop axial flow turbines.
 (ii) Annular and industrial type combustion chamber design (5)